

Nifty 100 Financial Intelligence Platform - Analyst Operations Guide

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This guide documents the production workflow for ingesting, validating, analysing, screening, and serving the Nifty 100 Financial Intelligence Platform.

System Architecture & Data Pipeline Overview

- ETL ingests SEC-style text files, normalises tickers and reporting years, and writes the relational warehouse to db/nifty100.db.
- The relational warehouse powers historical profit and loss, balance sheet, cash flow, document, stock price, and analytical ratio tables.
- The NLP layer parses unstructured disclosures, extracts CAGR-style phrases with regex, and generates heuristic pros and cons.
- The screener layer computes sector-aware quality scores, percentile comparisons, and clustering-ready analytical features.
- The FastAPI layer exposes company profiles, ratios, peer comparisons, screeners, health diagnostics, and clustering summaries under /api/v1/.

How to Navigate the Streamlit Dashboard

- Home: high-level KPI snapshots, sector composition, and cluster summary shortcuts.
- Profile: single-company deep dive, latest KPIs, time-series charts, and sentiment notes.
- Screener: preset investment filters, threshold sliders, and downloadable CSV outputs.
- Peers: benchmark radar views and percentile-based peer matrix comparisons.
- Trends: company-level historical overlays for up to three metrics at a time.
- Sectors: sector bubble charts and sector-median KPI views.
- Capital: capital allocation pattern map and company groupings.
- Reports: access compiled PDF report artifacts and reporting-year links.

Using the Quantitative Screener Engine

- Preset strategies such as Quality Compounder, Value Pick, Growth Accelerator, Dividend Champion, Debt-Free Blue Chip, and Turnaround Watch are available out of the box.
- Analysts can tune ROE, debt-to-equity, free cash flow, CAGR, OPM, P/E, P/B, dividend yield, and ICR constraints with sidebar sliders.
- The screener applies Financials-specific carve-outs so structurally leveraged sectors are not over-penalised.
- Filtered results can be exported directly to CSV for downstream modelling or portfolio review.

ReportLab PDF Generation Suite

Run the full institutional report batch with ``make sprint5-pdf-batch``.

- Company tearsheets: 92 two-page PDF reports with KPI cards, charts, and qualitative commentary.
- Sector reports: 11 sector synthesis packs covering group median KPIs and constituent matrices.
- Portfolio summary: a consolidated multi-page view of the full index with directional performance markers.

REST API Server Operations & Endpoint Reference

Start the server with ``make sprint6-api-start`` and monitor ``/api/v1/health`` for uptime and row-count diagnostics.

Route	Purpose
/api/v1/companies	Universe summary, optional search and sector filters
/api/v1/companies/{ticker}	Full company profile payload
/api/v1/companies/{ticker}/pl bs cashflow r ratios	Historical financial sub-arrays
/api/v1/screener	Filterable financial screener endpoint
/api/v1/peers/{group_name}	Peer percentile matrix export
/api/v1/clusters	Cluster labels, outliers, stats, and plot paths
/api/v1/health	Uptime and warehouse row counts

REST API Example Integrations

cURL Example

```
curl http://127.0.0.1:8000/api/v1/companies/RELIANCE/ratios?from_year=2020&to_year=2024
```

Python Requests Example

```
requests.get('http://127.0.0.1:8000/api/v1/clusters').json()
```

Typical JSON Schema

```
{'company_id': 1, 'cluster_id': 0, 'cluster_name': 'High-Quality Compounders', 'distance_from_centroid': 0.52}
```

Machine Learning Clustering & Profiling

- The clustering pipeline derives ROE, debt-to-equity, revenue CAGR, FCF CAGR, and OPM features for the latest reporting year.
- Missing values are imputed using broad-sector medians before the feature matrix is standardised.
- KMeans is executed with $k=5$ and $\text{random_state}=42$, then labelled into analyst-friendly archetypes.
- Entity-level Z-scores are computed within sector cohorts to flag outliers above an absolute threshold of 3.
- Pearson correlation matrices and elbow plots are exported to support model diagnostics and analyst interpretation.

Troubleshooting, Maintenance & Performance Tuning

- SQLite locks: close concurrent DB writers and re-run the affected make target after the prior process exits.
- Caching limits: clear Streamlit cache when stale dashboard values persist between major data refreshes.
- API reachability: confirm `make sprint6-api-start` is active before using dashboard endpoint shortcuts.
- Load testing: use lightweight batched reads and avoid simultaneous PDF generation during heavy API testing.

Appendix & Glossary of Financial KPIs

KPI	Definition
ROE	Net profit divided by equity capital plus reserves, expressed as a percentage.
ROCE	EBIT divided by capital employed.
NPM	Net profit divided by revenue.
OPM	Operating profit divided by revenue.
D/E	Borrowings divided by shareholder equity.
ICR	Operating profit plus other income divided by interest.
FCF Yield	Free cash flow divided by market capitalisation.
CAGR	Compound annual growth rate across a defined historical window.